Owain **West**

owest@sas.upenn.edu | owainwest.com | owainwest | owainwest

Senior engineer building verification platforms, compilers and data systems in Rust.

Experience

TickPick | Senior Data Engineer New York | July 2026 - Present

- Conceived and independently built **Eye**, a general-purpose Rust verification platform for humans and AI agents. It joins evidence across data, code, analytics, deployments and work tracking, showing what holds, what fails and what evidence is still missing.
- Built **Jig**, an OpenAPI/JSON Schema compiler generating checked Rust types, HTTP/MCP bindings and Python adapters; generated constructors enforce schema and domain constraints.
- Designed shared typed identities and a reusable judgment algebra separating source readers from pure checks. Composed answers preserve provenance, observation time and missing-evidence requirements. Coverage accounting distinguishes incomplete source reads from shortened displays, so a display limit cannot silently become a completeness claim.
- Implemented symbolic analysis of deployment rules, tracing which branch paths reach production and under what review protection, alongside exact-arithmetic reconciliation and downstream dependency checks.
- Drove data-team adoption of Eye and taught colleagues Rust, with extension guides for adding source readers.
- Built and deployed a Google Ads MCP service with Microsoft Entra OAuth and server-enforced tool authorization, giving AI assistants authenticated campaign analytics, bid simulations and historical warehouse reads through Cowork.

Beyond Identity | Data Engineer II New York | May 2024 - June 2026

- Standardized OpenAPI-driven service generation in Rust/Axum through a shared scaffolding workflow for new microservices.
- Built Rust/Kafka microservices with Tokio, Axum and rdkafka for Elasticsearch, Splunk, Datadog and Sentinel integrations.
- Introduced typed specifications for agentic development so AI-generated deliverables could be audited against explicit requirements, and drove adoption of the workflow.
- Implemented credential escrow for MCP tools in the **Ceros** agentic security harness.
- Delivered thread-safe, real-time Rust log and metric aggregation powering the customer-facing **SIEM Logs** feature.
- Developed real-time Go/Rust threat signals for zero-trust authentication, including IP novelty and travel-speed heuristics.
- Managed AWS infrastructure for Rust/Go services with Terraform, Kubernetes and GitLab CI/CD.

TickPick | Data Engineer New York | May 2022 - May 2024

- Led data engineering across Snowflake, Databricks and Azure ML, operating **400+ pipelines** loading hundreds of millions of rows monthly and mentoring a junior engineer.
- Automated acceptance of inbound ticket transfers, enabling launch of TickPick's **autotransfer** app feature.
- Automated daily pricing comparisons across **100k+ competitor listings**, supplying the competitive-pricing data behind TickPick's price-dominance strategy.
- Evaluated ETL vendors and negotiated contracts, reducing annual costs by **more than 70%** in one case.

Royal Bank of Canada, Toronto Innovation Lab | Software Engineer Summer Hire Toronto | May-Aug, 2017-2019

- Co-inventor of **US Patent 11,403,465 B2** for natural-language and image-processing methods in financial document analysis.
- Built a natural-language market-research interface and benefits-selection prototypes later refined and adopted company-wide.

University of Pennsylvania | Head Teaching Assistant; LGIC 010 Philadelphia | Jan-May, 2017-2019

- Coauthored the LGIC logic **textbook**; led the teaching-assistant team and contributed to curriculum development.

Projects

- **gargi** (private; walkthrough on request) — hand-written Lean 4 parser in Rust for a personal formalization project. Parses all of Mathlib (235k declarations) in about 4 s single-threaded, with no error recorded on 98.4% of declarations.
- **proth** (private; walkthrough on request) — Proth-prime search in Rust: segmented sieve, then Proth's test with squaring matched to each cache tier (Montgomery, three-prime NTT, Proth-shape reduction, AVX-512 batches of 16) and a Metal GPU backend.
- **sheaf** (public) — C/JavaScript esolang with an interpreter, compiler emitting C, and manuscript-review engine; includes diagnostics, executable examples and a browser playground.

Education

University of Waterloo | MMath | Computer Science Waterloo | Sep 2019 - Dec 2020

- Selected graduate coursework: Computer-Aided Verification; Automatic Sequences.

University of Pennsylvania | BA, Magna Cum Laude | Triple Major Philadelphia | Sep 2015 - May 2019

Logic, Information, and Computation; Computer Science; Philosophy and Science

- Selected graduate coursework: Logic and Computability; Finite Model Theory; Networked Systems.

Expertise

- **Languages and systems**: Rust, Go, Python, SQL, OCaml, Julia, Lean 4; Tokio, Axum, Kafka.
- **Developer infrastructure**: OpenAPI, JSON Schema, MCP, OAuth, Microsoft Entra, GitHub Actions, GitLab CI/CD; AWS, Azure, Terraform, Docker, Kubernetes.